Employee Attrition Analysis Using Exploratory Data Analysis and Machine Learning

Abstract

This educational project examines 1,470 employee records with 31 columns. 237 records are labelled as attrition, a dataset proportion of 16.1%. The project combines descriptive analysis with a reproducible Random Forest classification workflow. Dataset provenance was not supplied, so findings must not be attributed to a named company without verification.

1. Introduction

Employee attrition can affect organizational continuity, recruitment effort and institutional knowledge. Data analysis can reveal patterns associated with employee departures. This project demonstrates descriptive exploration and a baseline supervised-learning workflow.

2. Dataset and Method

The CSV contains 1,470 rows and 31 columns. Attrition is encoded as 1 for employees who left and 0 for employees who stayed. The file has 0 missing cells. The model uses a stratified 80/20 train-test split, median imputation for numeric columns, most-frequent imputation and one-hot encoding for categorical columns, followed by a class-weighted Random Forest classifier.

3. Descriptive Findings

Measure	Stayed (0)	Left (1)
Records	1,233	237
Average age	37.6	33.6
Average monthly income	6,833	4,787
Overtime = Yes	23.4%	53.6%

The attrition-labelled group has a lower average age and lower average monthly income than the retained group. Overtime is more common in the attrition-labelled group. These are descriptive associations and do not establish causation.

4. Machine Learning Design

The dashboard trains a Random Forest pipeline on demand and displays accuracy, precision, recall, a classification report and a confusion matrix. Metrics are calculated in the runtime and are not claimed as pre-verified performance. Evaluation on new, representative data is required before operational use.

5. Limitations and Responsible Use

The dataset source, sampling design, time period and organizational provenance were not supplied. Therefore, this report does not establish that the records belong to Palo Alto Networks or any specific employer. Predictive systems can reproduce historical bias. Attrition scores must not be used as the sole basis for hiring, promotion, performance or termination decisions. Further validation, fairness testing, privacy review and domain expertise are necessary.

6. Conclusion

The project demonstrates data validation, descriptive analysis, visualization and baseline classification in a reproducible Streamlit application. Separate Python modules improve maintainability and make the repository easier to review.

References

1. pandas documentation: https://pandas.pydata.org/docs/
2. scikit-learn documentation: https://scikit-learn.org/stable/
3. Streamlit documentation: https://docs.streamlit.io/